The Journal of Open Source Software

Pycashier: cash in on DNA barcode tags

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DOI: 10.xxxxxx/draft

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Submitted: 12 April 2024
Published: unpublished

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Summary

Pycashier is a tool designed to extract DNA barcode tags from next generation sequencing data. It was originally developed for use with the ClonMapper Barcoding System (Al'Khafaji et al., 2018; Gardner et al., 2022, 2024). Pycashier has been previously utilized to interrogate tumor heterogeneity in barcoded cancer cell-line models (Gutierrez et al., 2021; Johnson et al., 2020), however, it is generalizable to similar DNA barcoding systems with known flanking regions.

Statement of need

DNA sequencing and DNA barcoding specifically, have become more common as a modality for the characterization of clonal and lineage-specific subpopulations of cells. As researchers leverage these technologies, they'll require tools easy to setup and use to facilitate downstream biological analysis. DNA barcode sequencing suffers from several sources of noise that must be accounted for prior to statistical analysis. This noise can arise in typical Polymerase Chain Reaction (PCR) preparation (Kebschull & Zador, 2015; Potapov & Ong, 2017) or during read-out (Manley et al., 2016). Historically, the analysis of DNA barcoding has relied on tailored computational workflows, such as TimeMachine (Emert et al., 2021), that are difficult to parameterize or extend to similarly designed DNA barcoding systems. Recently, there has been the development of several NextFlow-based techniques, such as BARtab (Holze et al., 2024) and NextClone (Putri et al., 2023), which offer improved support for generalized approaches to processing barcode sequencing data. However, they require familiarity with NextFlow, which may be uncommon for experimentalists. Pycashier aims to be simple to install and generalizable enough to be useful to the broader community while also providing a user friendly interface.

Implementation and Usage

Pycashier was intentionally designed to be simple-to-use for both computational and experimental biologists. It accomplishes this by leveraging purpose-built software for an opinionated DNA barcode processing pipeline. Pycashier has a command-line interface (CLI) implemented in python. Users have the option of installing pycashier with pip, conda/mamba (from conda-forge), or as a standalone Docker image which includes all necessary runtime dependencies for maximum reproducibility. Pycashier maintains outputs and logs of all steps for simple debugging and reuse across a project. The pycashier CLI has four subcommands to facilitate processing of DNA barcode sequencing data. Users can specify parameters either through command-line flags or through a toml file.

Pycashier is primarily used for generating counts of individual barcode sequences from targeted PCR amplifications of DNA-barcoded cells. This is done with pycashier extract, which accepts a directory of fastq files directly from Illumina sequencing and generates a tsv of individual barcodes and counts for each input fastq. First, filtering is performed with fastp

(Chen et al., 2018) to remove low quality sequencing reads. Next, flanking sequences are used to extract a region of interest with `cutadapt` (Martin, 2011). Then, the list of identified sequences are corrected for errors introduced in either preparation or sequencing using a message passing clustering powered by `starcode` (Zorita et al., 2015). Finally, minimal count filtering is applied to remove any remaining noise from sequencing.

In addition to barcode extraction from targeted sequencing, `pycashier` facilitates barcode extraction from single-cell RNA-sequencing (scRNA-seq). Specifically, it's compatible with data generated with the 10X Genomics 3' based single cell gene expression kit. In this case, the command `pycashier scrna` accepts `sam` files derived from processed `10X` data and generates a `tsv` with cell/UMI resolved barcode sequences, which can then be mapped directly back to the transcriptome of individual cells. To accomplish this, `pycashier` first extracts cell/UMI and read sequences from `sam` files using `pysam` (*Pysam-Developers/Pysam*, 2024). Next, sequences are individually extracted from `fastq` files again, using `cutadapt` with known flanking sequences.

`Pycashier` provides two additional convenience commands: `merge`, to generate single read consensus sequences from paired-end sequencing, and `reciept`, to combine output `tsv` files from `pycashier extract` along with calculating some basic metrics across samples.

Documentation and further usage instructions for `pycashier` can be found at docs.brocklab.com/pycashier.

Acknowledgments

We acknowledge Aziz Al'Khafaji for the development of the ClonMapper Barcoding System, which drove the development of this tool. The authors are grateful for support from NIH 1U01CA25354 (to A.B).

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